LINUX ADMINISTRATION LAB SHEET — 1

Course Code: DCS583
Subject: Linux Administration

Q1. Print your current working directory to verify your location in the filesystem.

suhani@ubuntu:~$ pwd

Q2. List all files and directories in your current location using the basic ls command.

suhani@ubuntu:~$ ls

Q3. List all contents of your home directory, including hidden files.

suhani@ubuntu:~$ ls -a ~

Q4. Display all files in long-listing format with human-readable file sizes.

suhani@ubuntu:~$ ls -lh

Q5. Create a new directory named myfirstlab in your current directory.

suhani@ubuntu:~$ mkdir myfirstlab

Q6. Navigate into the newly created myfirstlab directory.

suhani@ubuntu:~$ cd myfirstlab

Q7. Inside myfirstlab, create three empty text files: a.txt, b.txt, and c.txt using a single command.

suhani@ubuntu:~/myfirstlab$ touch a.txt b.txt c.txt

Q8. Display the full contents of the system file /etc/os-release on your terminal screen.

suhani@ubuntu:~$ cat /etc/os-release

Q9. Check the username of the user currently logged into the terminal session.

suhani@ubuntu:~$ whoami

Q10. Display your system's hostname on the terminal.

suhani@ubuntu:~$ hostname

Q11. Check how much disk space is currently free on your system in human-readable format.

suhani@ubuntu:~$ df -h

Q12. Check the total system RAM and current memory usage in human-readable units.

suhani@ubuntu:~$ free -h

Q13. Check how long your Linux system has been continuously running.

suhani@ubuntu:~$ uptime

Q14. Display full system and kernel operating information using uname.

suhani@ubuntu:~$ uname -a

Q15. Use the manual pages to look up the documentation for the ls command.

suhani@ubuntu:~$ man ls

Press q to exit the manual page.

Q16. Display a short quick-usage summary for the cp command using the --help flag.

suhani@ubuntu:~$ cp --help

Q17. Use the whatis command to get a one-line description of the mv command.

suhani@ubuntu:~$ whatis mv

Q18. Navigate back to your home directory using the shorthand tilde (~) symbol.

suhani@ubuntu:~$ cd ~

Q19. Create a nested folder hierarchy project/src/data in a single command using mkdir.

suhani@ubuntu:~$ mkdir -p project/src/data

Q20. Delete the file c.txt created inside the myfirstlab directory.

suhani@ubuntu:~$ rm ~/myfirstlab/c.txt

Q21. Create a file named students.txt and add five student names, one per line, using echo and the append redirection operator (>>).

suhani@ubuntu:~$ echo "Aman" > students.txt
suhani@ubuntu:~$ echo "Riya" >> students.txt
suhani@ubuntu:~$ echo "Rahul" >> students.txt
suhani@ubuntu:~$ echo "Priya" >> students.txt
suhani@ubuntu:~$ echo "Suhani" >> students.txt

Q22. Display only the first two names from students.txt using the head command.

suhani@ubuntu:~$ head -n 2 students.txt

Q23. Display only the last two names from students.txt using the tail command.

suhani@ubuntu:~$ tail -n 2 students.txt

Q24. Create a directory named backup and copy students.txt into it without navigating out of your current directory.

suhani@ubuntu:~$ mkdir -p backup
suhani@ubuntu:~$ cp students.txt backup/

Q25. Rename students.txt to class_list.txt using the mv command.

suhani@ubuntu:~$ mv students.txt class_list.txt

Q26. Use grep to search inside class_list.txt for any name containing the letter 'a'.

suhani@ubuntu:~$ grep -i "a" class_list.txt

Q27. Display line numbers along with matching text when searching for a string in class_list.txt using grep -n.

suhani@ubuntu:~$ grep -n "a" class_list.txt

Q28. Change permissions of class_list.txt so only the owner has read and write rights, with no execute permission, and verify using long listing.

suhani@ubuntu:~$ chmod 600 class_list.txt
suhani@ubuntu:~$ ls -l class_list.txt

Q29. Find all files ending with .txt in your home directory tree using the find command.

suhani@ubuntu:~$ find ~ -type f -name "*.txt"

Q30. Use wc -l to count the exact number of lines (students) in class_list.txt.

suhani@ubuntu:~$ wc -l class_list.txt

Q31. Display the first five lines of /etc/passwd on the terminal.

suhani@ubuntu:~$ head -n 5 /etc/passwd

Q32. Display the last five lines of /var/log/dpkg.log or /var/log/syslog.

suhani@ubuntu:~$ tail -n 5 /var/log/dpkg.log

Alternative command:

suhani@ubuntu:~$ tail -n 5 /var/log/syslog

Q33. Copy an entire directory structure (project/) into a new folder named project_backup/ recursively.

suhani@ubuntu:~$ cp -r project/ project_backup/

Q34. View a large text file page-by-page using less and identify the key used to quit the pager.

suhani@ubuntu:~$ less class_list.txt

Press q to exit.

Q35. Move two files (a.txt and b.txt) simultaneously into a target folder named archive/.

suhani@ubuntu:~$ mkdir -p archive
suhani@ubuntu:~$ mv ~/myfirstlab/a.txt ~/myfirstlab/b.txt archive/

Q36. Create a directory named logs, create app.log inside it with ten lines of dummy text, and then view the last three lines using tail.

suhani@ubuntu:~$ mkdir -p logs
suhani@ubuntu:~$ for i in {1..10}; do echo "Dummy log line $i" >> logs/app.log; done
suhani@ubuntu:~$ tail -n 3 logs/app.log

Q37. Use find paired with -exec to locate all .log files in your current working path and display their contents in a single command line.

suhani@ubuntu:~$ find . -type f -name "*.log" -exec cat {} ;

Q38. Recursively search all files in your current directory tree for the word 'error' using grep -r.

suhani@ubuntu:~$ grep -ri "error" .

Q39. Create a shell script file test.sh, grant it execute permissions using symbolic chmod syntax (+x), and write a comment inside explaining the permission string rwxr-xr--.

suhani@ubuntu:~$ printf '#!/bin/bash\n# rwxr-xr-- means owner: read/write/execute, group: read/execute, others: read only.\necho "Test script"\n' > test.sh
suhani@ubuntu:~$ chmod +x test.sh
suhani@ubuntu:~$ ls -l test.sh

Q40. Write a single-line command using chaining (&&) to create a folder demo, enter it, and list its contents.

suhani@ubuntu:~$ mkdir -p demo && cd demo && ls

Q41. Combine find and wc -l in a single pipeline to count how many total .txt files exist under your home directory tree.

suhani@ubuntu:~$ find ~ -type f -name "*.txt" | wc -l

Q42. Find all directories (not regular files) inside /var/log using find -type d.

suhani@ubuntu:~$ find /var/log -type d

Q43. Search for a specific file system configuration entry inside /etc/fstab using grep, ignoring case sensitivity.

suhani@ubuntu:~$ grep -i "UUID" /etc/fstab

Q44. Check the disk space consumption of a specific folder in human-readable format using du -sh.

suhani@ubuntu:~$ du -sh ~/myfirstlab

Q45. Write a command that displays the three largest files in a directory by sorting file sizes.

suhani@ubuntu:~$ find . -maxdepth 1 -type f -printf '%s %p\n' | sort -nr | head -n 3

Q46. Explain why running rm -rf * in an unverified path is hazardous, and state the safety verification command habit that prevents accidental deletion.

Command 1: Check the current working directory.

suhani@ubuntu:~$ pwd

Command 2: Verify the files and directories before deletion.

suhani@ubuntu:~$ ls -la

Q47. Use touch on an existing file to update its access and modification timestamps to the current system time without modifying its contents.

suhani@ubuntu:~$ touch class_list.txt

Q48. Create a file containing multiple duplicate lines, then use sort and uniq to display only unique lines from the file.

suhani@ubuntu:~$ printf "apple\nbanana\napple\norange\nbanana\n" > fruits.txt
suhani@ubuntu:~$ sort fruits.txt | uniq

Q49. Redirect the standard output of a command (e.g., uname -a) to a file named sys_info.txt, then view its content.

suhani@ubuntu:~$ uname -a > sys_info.txt
suhani@ubuntu:~$ cat sys_info.txt

Q50. Find all files in /var/log that were modified within the last seven days using find -mtime.

suhani@ubuntu:~$ find /var/log -type f -mtime -7